

Bank Customer Churn Analysis Dashboard

Customer Retention & Predictive Analytics
Executive Presentation

Business Problem

- Customer churn impacts revenue and profitability.
- Lack of proactive churn identification.
- Need for data-driven retention strategies.

Project Objectives

- Predict customer churn risk.
- Identify key churn drivers.
- Support targeted retention campaigns.

Dataset Overview

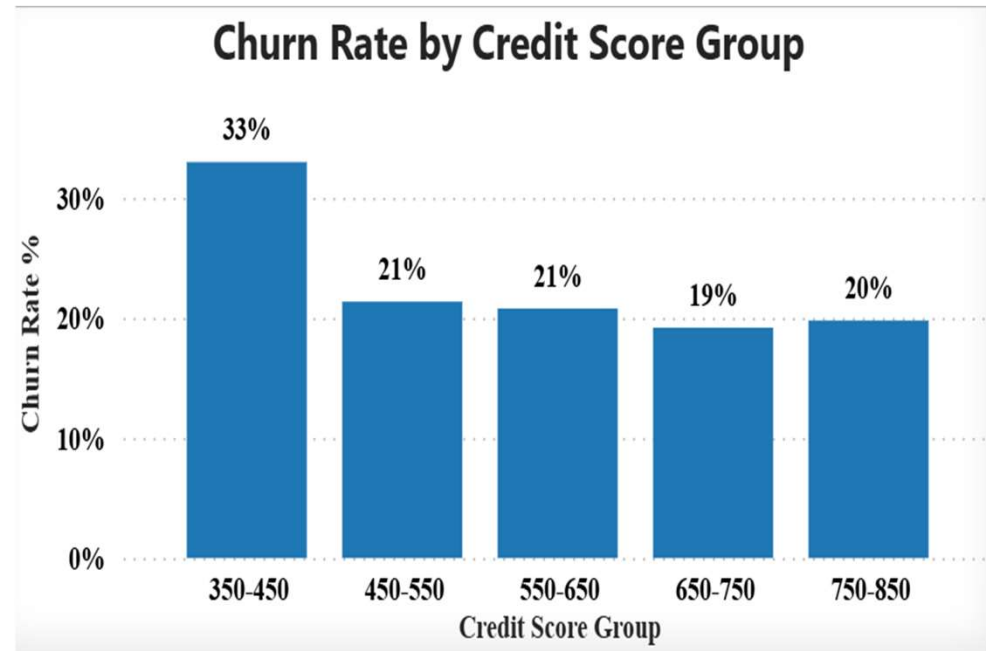
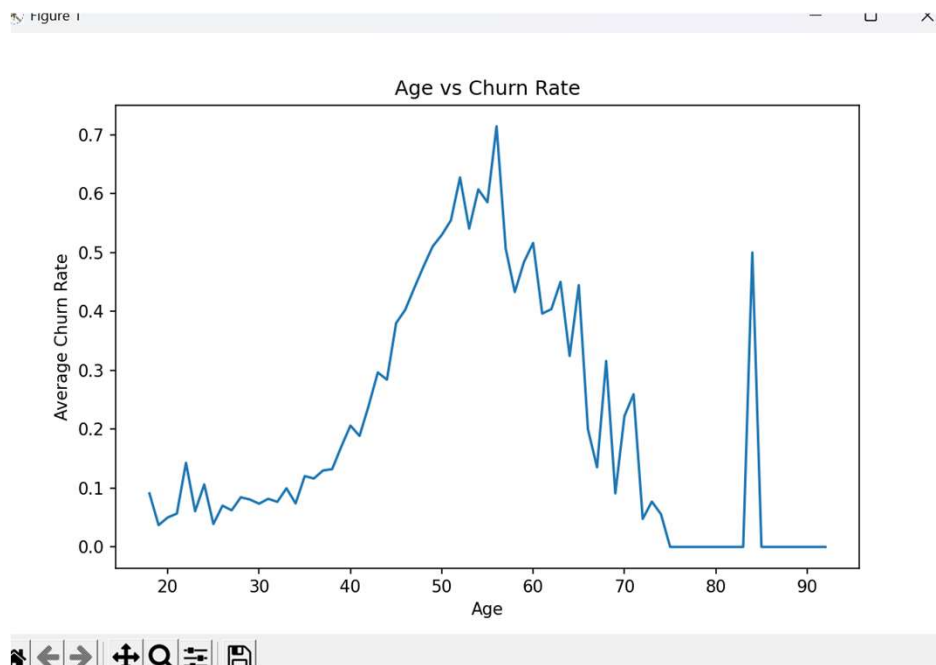
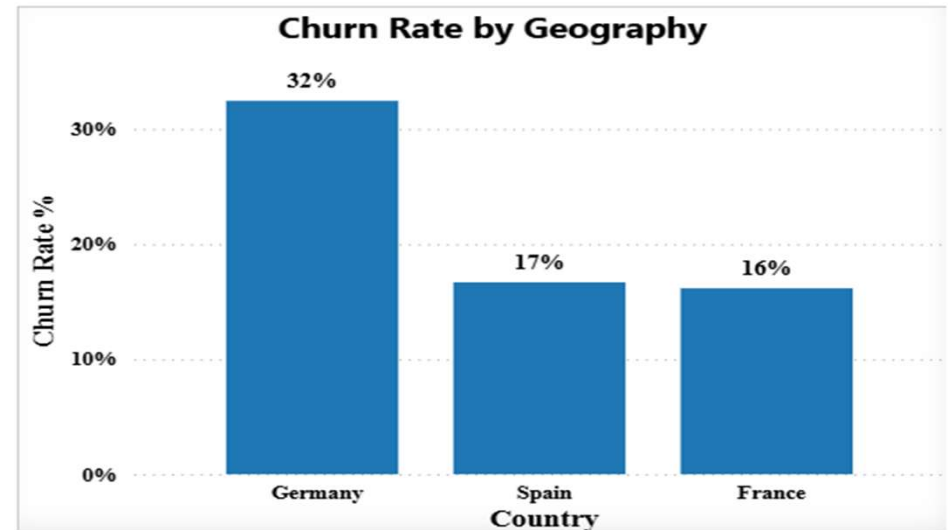
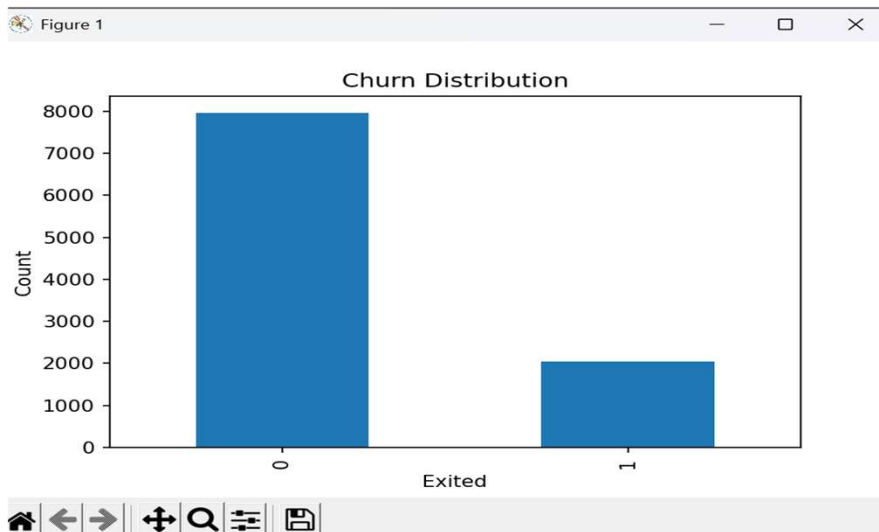
- 10,000 customers
- 16 features
- Demographic, financial, and engagement data.

Analytical Approach

Data Collection

- Preprocessing
- Feature Engineering
- Machine Learning Models
- Dashboard & Deployment

Exploratory Data Analysis



Key Churn Drivers

- Product Density
- Age
- Credit Score
- Estimated Salary
- Balance

Model Performance

Random Forest:

Accuracy: 86.35%

ROC-AUC: 0.8453

Predictive Application

Insert Streamlit screenshots here:

- Churn Probability
- Risk Level
- Stay/Leave Prediction

Business Recommendations

- Target high-risk customers.
- Improve product engagement.
- Personalized retention campaigns.
- Focus on inactive customers.

Business Impact

- Proactive retention strategy.
- Improved customer lifetime value.
- Reduced churn-related revenue loss.

Thank You